Response Function for Generalized Heterodyne Optical-Sampling Techniques (RF GHOST)

User manual

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System requirements

RF GHOST general system requirements are Linux/Unix machines with gfortran/f95 installed. Any text editor (vi, gimp etc.) will be suitable to edit the input files. Fast fourier transform libraries are provided with the RF GHOST.

Warranty

The PIGLET comes with absolutely no warranty. Authors are not responsible for whatever damage this program might cause. Efforts were made to make the program stable. However, it is definitely possible to choose a set of parameters for which the program will give wrong results, senseless results, or simply crash. Treat this program not as a foolproof software, but more as an experimental setup, which can give all kinds of results depending on handling.

License agreement

Setup

1. Unzip the installation archive (tar -xf rh_ghost.tar) in your working directory.

2. Compile the fourier transform libraries, using the following commands:

f95 -c -o cfftb.o -fallow-argument-mismatch cfftb.f
f95 -c -o cfftf_cut.o -fallow-argument-mismatch cfftf_cut.f
f95 -c -o fourier.o -fallow-argument-mismatch fourier.f

The execution of the first two commands can look like this_

```
gusakov@alcazaba:package$ f95 -c -o cfftb.o -fallow-argument-mismatch cfftb.f
cfftb.f:207:72:

  207 |       IF (J-4) 102,102,103
      |                                                                        1
Warning: Fortran 2018 deleted feature: Arithmetic IF statement at (1)
cfftb.f:213:72:

  213 |       IF (NR) 101,105,101
      |                                                                        1
Warning: Fortran 2018 deleted feature: Arithmetic IF statement at (1)
cfftb.f:74:72:

   74 |       CALL CFFTB1 (N,C,WSAVE,WSAVE(IW1),WSAVE(IW2))
      |                                                                        1
Warning: Type mismatch in argument 'ifac' at (1); passed REAL(4) to INTEGER(4)
cfftb.f:192:72:

  192 |       CALL CFFTI1 (N,WSAVE(IW1),WSAVE(IW2))
      |                                                                        1
Warning: Type mismatch in argument 'ifac' at (1); passed REAL(4) to INTEGER(4)
gusakov@alcazaba:package$ f95 -c -o cfftf_cut.o -fallow-argument-mismatch cfftf_cut.f
cfftf_cut.f:63:72:

   63 |       CALL CFFTF1 (N,C,WSAVE,WSAVE(IW1),WSAVE(IW2))
      |                                                                        1
Warning: Type mismatch in argument 'ifac' at (1); passed REAL(4) to INTEGER(4)
gusakov@alcazaba:package$
```

3. Compile and link the main code, using

f95 rf_ghost.f cfftb.o cfftf_cut.o fourier.o

4. To run the program, run a.out (./a.out).

Compiling the main code and running it result in the following output:

```
gusakov@alcazaba:package$ f95 rf_ghost.f cfftb.o cfftf_cut.o fourier.o
gusakov@alcazaba:package$ ./a.out
 nt_pad       64032
 Nz=           10  Nceo=            1 Nt=        16008 Ndel=         4002
 dn =  -1.3789655703293491E-002
 corr    2000000000.0000000       0.27343750000000000       0.27343750000000000
           1  -0.0000000000000000
   1.0000000000000000
 ch -0.17491339469538664      -5.8304464898462201E-002  -1.2183908045977014E-023  -1.2183908045977014E-023
^C
```

Input files

Input files are:
input.dat, sampling_wavefcorm_r.dat, sampling_waveform_i.dat, filter_resp.dat, diode_resp.dat

Input files must contain the following number of lines:

input.dat	17
sampling_waveform_r.dat	20001
filter_resp.dat	10000
diode_resp.dat	10000

All the parameters must be set, there are no default values. Including less than above-indicated number of lines will produce an error; including more lines will lead to excess lines being ignored. Each of the lines should contain the required value(s) in the first position(s), separated by spaces, followed by an optional comment which will be ignored. Almost all values, with an exception of the ones defined otherwise below, are defined in SI units. In the package, the files used to generate Fig. 5(b) of Ref. [1] are provided as an example. The parameters can be given using the floating-point notation: e.g. 5d-5 means to 5×10^{-5}. The meaning of the parameters in the context of the simulation is provided and explained in detail in Ref. [1].

input.dat

File input.dat must contain 5 lines which define various parameters related to input pulses, materials, photoionization model, and so on, as shown in the following table:

Line number	Integer/real	Units	Meaning
1	real	s	Time step
2	real	s	time window
3	real	-	Minimum value of CEP of the sampling pulse
4	real	-	Step of CEP of the sampling pulse
5	real	-	Maximum value of CEP of the sampling pulse
6	integer	-	A number which selects the ionization model. Negative value -N: multiphoton model with N as number of photons 0: Ivanov-Yudin model 3: ADK model 1: a model which combines cycle-averaged ionization rate of Ivanov-Yudin model and subcycle ionization rate of ADK model
7	integer	-	0/1: 1 switches the Brunel radiation on, 0 switches the Brunel radiation off
8	integer	-	0/1: 1 switches the injection on, 0 switches the injection off
9	integer	-	0/1: 1 switches the second-order susceptibility on, 0 switches the second-order susceptibility off

10	integer	-	0/1: 1 switches the third-order susceptibility on, 0 switches the third-order susceptibility off
11	real	W/m^2 or V/m	Strength of the sampling pulse. If positive, defines the peak intensity; if negative, defines the peak electric field strength
12	real	m	Step of the propagation coordinate
13	real	m	Thickness of the sample
14	real	-	Minimum strength of the sampling pulse, relative to the value defined in line 11. The minimum value is used for minimum-CEP case.
15	real	-	Maximum strength of the sampling pulse, relative to the value defined in line 11. The maximum value is used for maximum-CEP case.
16	integer	-	A flag which determines whether the auxiliary field which is required for separation of different Kerr contributions (same-sign, different-signs) is calculated, as discussed in Section 3 of Supplementary materials of Ref. [1]. 1 switches the calculation on, 0 switches the calculation off. This flag should be set to 1 only is the next flag set to non-zero, no avoid unnecessary numerical effort.
17	integer	-	0: include both same-sign and different-sign contributions of Kerr nonlinearity 1: include only different-sign contributions of Kerr nonlinearity 2: include only same-sign contributions of Kerr nonlinearity

Note that depending on the values of parameters 3,4, and 5, several calculations are performed, with CEP of the sampling file linearly varying from the minimum value to the maximum value with the step defined in line 4. Simultaneously and also linearly, the maximum value of the sampling pulse electric field varies from the value in line 14 to the value in line 15.

This is a typical example of input.dat:

```
2.500000E-17
5.000000E-14
0.000000E+00
1.500000E-03
1.100000E-03
-8
1.000000E+00
1.000000E+00
0.000000E+00
1.000000E+00
-4E+10
20E-10
20E-9
1.000000E+00
1.000000E+00
0
0
```

sampling_waveform_i.dat and sampling_waveform_r.dat

These files are the real and imaginary parts of the sampling pulse waveform. Both files consist of 20001 lines, providing relative field values at time grid with time step of 0.02 fs. The maximum of the provided values, calculated over both files, should be equal to 1.

filter_resp.dat and diode_resp.dat

These files are the transmission of the filter and the photodiode. Both files consist of 10000 lines, which denote the corresponding transmission at frequencies starting from 0.78892752 fs^{-1}, with a step of 1.15492×10^{-5} fs^{-1}.

In addition, the parameters of the material are set in the code of rh_ghost.dat at lines 144-150 and its chromatic dispersion is set at lines 168-170.

Output files

The response file(s) are named as results_cepN.dat, where N is the case number, as defined by the CEP minimum and maximum values and CEP step, as detailed above. Each results_cepN.dat file contains 2000 lines. In each line

first value is frequency in units of PHz
second value is absolute value of the response function, as defined by Eq. (3) of Ref. [1], in arbitrary units
third value is the phase of the response function, dimensionless.

Besides that, several other files are written at the output. These files are not relevant in the context of the response function calculation.

References

[1] https://arxiv.org/abs/2410.19196